ModelIt!

Teacher's Guide

Who Eats Who? Energy Flow Through Food Webs

Tracing Energy from Sunlight to Top Predators

Quick Reference

Grade Level	Grade 06L2
NGSS Standard	MS-LS2-3, MS-LS2-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS2-3, MS-LS2-4

Develop a model to describe the cycling of matter and flow of energy among living and nonliving parts of an ecosystem. Construct an argument supported by empirical evidence that changes to physical or biological components of an ecosystem affect populations.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how solar energy flows through trophic levels with 90% loss at each transfer, and how decomposers recycle matter.
Disciplinary Core Idea	**LS2.B Cycles of Matter and Energy Transfer in Ecosystems** — Food webs are models that demonstrate how matter and energy are transferred between producers, consumers, and decomposers as the three groups interact within an ecosystem.
Crosscutting Concept	**Energy and Matter: Flows, Cycles, and Conservation** — Students trace energy flow (one-way, from Sun through organisms as heat) and matter cycling (nutrients recycled by decomposers) through the ecosystem.

CAST Testing Alignment

MULTIPLE CHOICE:

In a grassland ecosystem, producers capture 20,000 kJ of solar energy. Based on the 10% energy transfer rule, approximately how much energy is available to secondary consumers?

CONSTRUCTED RESPONSE:

Wolves were reintroduced to Yellowstone National Park in 1995. Using your understanding of trophic cascades and energy flow, explain how adding wolves affected the elk population, the vegetation, and even the rivers.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Food webs are the interconnected feeding relationships that link every organism in an ecosystem. Understanding energy flow through these webs reveals why ecosystems are structured the way they are and why every species matters.

Energy Flow Is One-Way

Unlike nutrients, energy does not cycle through ecosystems. Solar energy enters through photosynthesis, flows through consumers, and exits as heat. Every metabolic process (breathing, moving, growing) converts usable chemical energy into heat that radiates away. This one-way flow means ecosystems require constant solar input to function.

The 10% Rule

On average, only about 10% of the energy at one trophic level is stored as biomass at the next level. The other 90% is used for metabolic processes and lost as heat. This is why a field of grass can support fewer rabbits, which support even fewer foxes. It takes roughly 10,000 kg of grass to produce 1,000 kg of rabbits to produce 100 kg of foxes.

Trophic Cascades

When a predator population changes, it cascades through the food web. In Yellowstone, removing wolves led to an elk population explosion, which overgrazed willows and aspens along riverbanks, which caused riverbank erosion. Reintroducing wolves in 1995 reversed this cascade, allowing vegetation to recover and even changing how rivers flowed.

Decomposers Close the Loop

While energy flows one-way, matter cycles. Decomposers (bacteria, fungi, worms) break down dead organisms and waste, releasing nutrients like nitrogen and phosphorus back into the soil. Producers absorb these nutrients to grow, starting the cycle again. Without decomposers, nutrients would be locked in dead matter and ecosystems would collapse.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify solar energy input, producer biomass, primary consumer biomass, secondary consumer biomass, available energy, and decomposer return rate as the key components of the food web energy system.

E **Establish Relationships**
Students determine that solar energy positively drives producer biomass, each trophic level positively feeds the next but with 90% energy loss, and decomposers positively recycle nutrients back to producers.

V **Visualize & Model**
Students build a model showing energy flow from sunlight through three trophic levels with metabolic losses and nutrient recycling by decomposers.

E **Evaluate Outcomes**
Students run solar shutdown and predator removal scenarios to observe bottom-up and top-down cascade effects on the entire food web.

R **Revise & Extend**
Students add tertiary consumers or competition intensity to explore more complex food web dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Lions are the kings of the jungle, but there are only about 20,000 wild lions left. There are 7 billion chickens. Why?"*

Watch For

Show comparison images of lion vs. chicken populations. Let students wonder about the numbers.

Slide 2: Learning Objectives

3 min

Say: *"Today we're going to discover the hidden math that controls every ecosystem on Earth."*

Watch For

Have students read objectives aloud. Pre-teach 'trophic level' and 'biomass.'

Slide 3: Big Question

3 min

Say: *"The Sun blasts Earth with enough energy to power everything. So why can't ecosystems support more predators?"*

Watch For

Think-pair-share: Why are there more plants than animals in every ecosystem?

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model to trace every unit of energy from sunlight to the top predator."*

Watch For

Preview LEVER steps. Explain that this model has 6 components and more complex relationships.

Slide 5: Activity 1: Components

8 min

Say: *"Where does the energy start? What organisms capture it? Where does it go next?"*

Watch For

Guide sorting. Explain why Solar Energy Input is external (comes from outside the ecosystem).

Slide 6: Activity 2: Connections

8 min

Say: *"Here is the big surprise: 90% of energy is LOST at every step. Only 10% moves up."*

Watch For

Use a hands-on demo: start with 100 candies (energy). Pass 10 to herbivores, then 1 to predators. Where did the other 89 go?

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's cut the sunlight and watch the whole web collapse from the bottom up!"*

Watch For

Have students predict first. Run solar shutdown then predator removal. Compare cascade directions.

Slide 8: Discoveries

5 min

Say: *"Now you know the real reason there are so few eagles, sharks, and lions. It is all about energy math."*

Watch For

Connect the 10% rule to the energy pyramid shape. Discuss Yellowstone wolves as a real example.

Slide 9: STEM Challenge

12 min

Say: *"Can you design a sealed ecosystem that runs on nothing but light? That is what Mars colonists will need."*

Watch For

Distribute materials. Groups design terrariums with the right ratio of producers to consumers.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Solar Energy Input
- **INTERNAL:** Producer Biomass, Primary Consumer Biomass, Secondary Consumer Biomass, Available Energy at Each Level, Decomposer Return Rate

Relationships

- **Solar Energy Input to Producer Biomass:** POSITIVE (+) — More sunlight allows producers to photosynthesize more, building more biomass. Without sunlight, producers cannot make food and the entire web starves.
- **Producer Biomass to Primary Consumer Biomass:** POSITIVE (+) — More plant biomass supports more herbivores. But only about 10% of the energy in producers transfers to primary consumers; 90% is lost as heat.
- **Primary Consumer Biomass to Secondary Consumer Biomass:** POSITIVE (+) — More herbivore biomass supports more predators. Again, only 10% of herbivore energy reaches predators. This is why predators are rare.
- **Each Trophic Level to Available Energy:** NEGATIVE (-) — At each level, organisms use 90% of consumed energy for metabolism (breathing, moving, body heat). Only 10% is stored as biomass available to the next level.
- **Decomposer Return Rate to Producer Biomass:** POSITIVE (+) — Faster decomposition releases nutrients back into the soil sooner, allowing producers to grow more. Decomposers close the matter cycle.

Key Discoveries

- Only about 10% of energy transfers from one trophic level to the next. 90% is lost as heat from metabolism
- Removing top predators causes primary consumers to explode in population, which overgrazes producers
- Decomposers are essential because they recycle nutrients that producers need to capture new solar energy
- The energy pyramid explains why there are more plants than herbivores and more herbivores than predators
- Ecosystems depend on a constant input of solar energy because energy flows through but does not cycle

Multiple Choice Post-Assessment Answer Key

1. In a grassland food web, producers capture 10,000 kJ of solar energy. Using the 10% energy transfer rule, how much energy is available to secondary consumers?

Correct: C) 100 kJ because only 10% transfers at each step, and secondary consumers are two steps above producers

Correct! Producers capture 10,000 kJ. Primary consumers get 10% of that (1,000 kJ). Secondary consumers get 10% of the primary consumers' energy (100 kJ). Each trophic level retains only 10% of the energy from the level below.

2. In the food web model, what happens when secondary consumers are removed from the ecosystem?

Correct: B) Primary consumer biomass increases because nothing is eating them, which then causes producer biomass to crash from overgrazing

Correct! This is a top-down trophic cascade. Without predators, herbivore populations explode. The unchecked herbivores then overgraze the producers, causing producer biomass to crash. Removing predators destabilizes the entire web.

3. Which statement best explains why energy flows through an ecosystem but does not cycle?

Correct: B) Energy is converted to heat at every trophic level through metabolism, and heat radiates away and cannot be recaptured by organisms

Correct! At every trophic level, organisms use energy for metabolism (breathing, moving, maintaining body heat), and that energy is released as heat. Heat energy disperses into the environment and cannot be recaptured by living organisms, so new solar energy must constantly enter the system.

4. A marine biologist notices that removing sea otters from a kelp forest causes the kelp to disappear. Which explanation is supported by the food web model?

Correct: B) Without otters eating sea urchins, urchin populations exploded and devoured the kelp

Correct! This is a real trophic cascade. Sea otters are secondary consumers that eat sea urchins (primary consumers). Without otters controlling urchin populations, urchins multiply and overgraze the kelp (producers), destroying the entire kelp forest ecosystem.

5. Why are decomposers essential to ecosystem survival even though they are not part of the energy pyramid?

Correct: B) Decomposers recycle nutrients locked in dead matter back into the soil so producers can use them to grow

Correct! While energy flows one way and cannot be recycled, matter CAN cycle. Decomposers break down dead organisms and return essential nutrients (nitrogen, phosphorus, minerals) to the soil. Without decomposers, these nutrients would be permanently locked in dead matter and producers would run out of raw materials.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Energy recycles through ecosystems like nutrients do"	Energy does NOT cycle. It flows one way: in from the Sun, through organisms, and out as heat. Every metabolic process converts usable energy into heat that radiates into space. That is why ecosystems need constant sunlight. Nutrients cycle, but energy flows.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Removing predators helps prey populations"	In the short term, prey populations increase. But without predators, herbivores overpopulate and overgraze, destroying the producer base. Then herbivores starve in a crash. Predators actually stabilize prey populations at a healthy, sustainable level.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Decomposers are not important to the food web"	Decomposers are arguably the most critical organisms in any ecosystem. Without them, nutrients would be permanently locked in dead matter and producers would starve. They do not just clean up; they recycle the raw materials that keep the entire ecosystem running.